# Giri's Tech Hub, Pune.

## Core Java Machine Test

Batch : Jan/Feb/March-26

Date : 28/10/2026
Time : 08 to 10.00 Am.

**Instructions:**

1. Solve all questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

**Q1. Write a java program to print in this pattern.**

```
A                                   A
A   B                           B   A
A       C                   C       A
A           D           D           A
A   B   C   D   E   E   D   C   B   A
A           D           D           A
A       C                   C       A
A   B                           B   A
A                                   A
```

**Q2. Write a program to any languages find and display all contiguous subarrays whose sum equals a given target value. The array may contain positive, negative, and zero values.**
**Input: arr = [1, 2, 3, -2, 5, -3, 2]**
**target = 5**
**Output:**

**[2, 3]**
**[2, 3, -2, 5, -3]**
**[3, -2, 5, -3, 2]**
**[5]**

**Q3. Write a program to any languages find the longest contiguous palindromic substring from a given string without using any built-in reverse method.**
**Input: str = "forgeeksskeegfor"**
**Output: geeksskeeg**

**Q4. Write a Java program to group anagrams from a given array of strings using a Map.**
**Input:**
**String[] arr = {"eat", "tea", "tan", "ate", "nat", "bat"};**
**Output:**
**[[eat, tea, ate], [tan, nat], [bat]]**

**Constraint: Use a Map to group anagrams. Implement the logic manually without using built-in sorting methods, Streams, or predefined functions to identify anagrams. The order of groups may vary.**

**Q5. Write a Java program to find the contiguous subarray having the maximum sum from an ArrayList<Integer> in O(n) time complexity.**
**Input:**
**ArrayList<Integer> list = new ArrayList<>();**
**// Elements: -2, 1, -3, 4, -1, 2, 1, -5, 4**
**Output:**
**Maximum Sum: 6**
**Constraint:**

- **Implement the logic manually using Kadane's Algorithm.**
- **Do not use built-in methods such as Collections.max(), stream(), or other methods that directly calculate the result.**
- **Do not use nested loops.**
- **The solution must handle arrays containing all negative numbers.**
- **Use ArrayList to store the input elements.**

**Q6. Create an Employee class containing id, name, department, and salary.**
**Given a list of employees, sort the list using the following rules:**

1. **Sort by salary in descending order.**
2. **If salaries are equal, sort by name in ascending alphabetical order.**
3. **If both salary and name are equal, sort by employee ID in ascending order.**

**Requirements:**

- **Implement Comparable<Employee> for sorting by employee ID.**
- **Use a Comparator<Employee> for the required multi-level sorting.**
- **Demonstrate the difference between Comparable and Comparator.**
- **Display the employee list before and after sorting.**

**-----ALL THE BEST-----**